

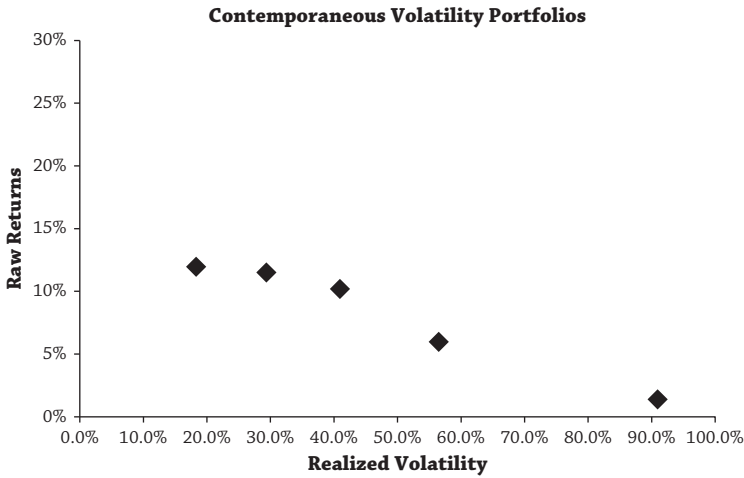


Figure 10.9

plots the average realized volatility and average realized returns of these quintile portfolios; there is still a negative relation between contemporaneous volatility and returns. Thus, the most volatile stocks *currently* lose money (which we cannot forecast), and they also tend to lose money in *the future* as well (which is predictable).

4.3 BETA ANOMALY

The first tests of the CAPM done in the 1970s did find positive relations between beta and expected returns, but they did not find that pure forms of the CAPM worked. Black, Jensen, and Scholes (1972), for example, found the relation between beta and returns to be “too flat” compared with what the CAPM predicted, but at least the relation was positive.

Fama and French wrote a major paper in 1992 that struck at the heart of the CAPM. While their main results showed that size and value effects dominated beta in individual stocks, they noted that “beta shows no power to explain average returns.” In fact, their estimated relation between beta and returns was statistically insignificant. Worse, the point estimates indicated that the relation between beta and returns was negative.

Lagged Beta and Future Returns

In Figure 10.10, I form quintile portfolios rebalancing every quarter based on betas estimated over the previous quarter using daily returns. The portfolios are equal weighted so as to form the largest differences in returns and Sharpe ratios, and returns are at the monthly frequency.

The beta anomaly is that stocks with high betas tend to have lower risk-adjusted returns. Panel A of Figure 10.10 shows that the average returns across